

Data Mining Chapter 12 Unsupervised Learning

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Question 4

Suppose that for a particular data set, we perform hierarchical clustering using single linkage and using complete linkage. We obtain two dendrograms.

- (a) At a certain point on the single linkage dendrogram, the clusters $\{1, 2, 3\}$ and $\{4, 5\}$ fuse. On the complete linkage dendrogram, the clusters $\{1, 2, 3\}$ and $\{4, 5\}$ also fuse at a certain point. Which fusion will occur higher on the tree, or will they fuse at the same height, or is there not enough information to tell?

Answer

In the case of complete linkage, the fusion will occur higher on the tree as Complete linkage depends on the maximum intercluster dissimilarity as the dissimilarity of the group.

- (b) At a certain point on the single linkage dendrogram, the clusters $\{5\}$ and $\{6\}$ fuse. On the complete linkage dendrogram, the clusters $\{5\}$ and $\{6\}$ also fuse at a certain point. Which fusion will occur higher on the tree, or will they fuse at the same height, or is there not enough information to tell?

Answer

With two points, the fusion will occur at the same height as the minimal (for single) and maximal (for complete) intercluster distance will be same.

Question 11

Write a Python function to perform matrix completion as in Algorithm 12.1, and as outlined in Section 12.5.2. In each iteration, the function should keep track of the relative error, as well as the iteration count. Iterations should continue until the relative error is small enough or until some maximum number of iterations is reached (set a default value for this maximum number). Furthermore, there should be an option to print out the progress in each iteration.

Test your function on the Boston data. First, standardize the features to have mean zero and standard deviation one using the `StandardScaler()` function. Run an experiment where you randomly leave out an increasing (and nested) number of observations from 5% to 30%, in steps of 5%. Apply Algorithm 12.1 with $M = 1, 2, \dots, 8$.

Display the approximation error as a function of the fraction of observations that are missing, and the value of M , averaged over 10 repetitions of the experiment.

Answer

To handle the missing values, an iterative PCA-based imputation procedure was implemented. The main idea is to replace the missing values initially with the mean of their corresponding variables and then repeatedly reconstruct the data matrix using a selected number of principal components.

The PCA reconstruction is obtained using the singular value decomposition (SVD):

$$X = PDQ^T.$$

For a chosen number of principal components M , the reconstructed matrix is

$$\hat{X}_M = P_{[:,1:M]} D_{1:M,1:M} Q_{[1:M,:] }.$$

The missing observations are then replaced by the corresponding values from this reconstructed matrix. This procedure is repeated until the relative change in the reconstruction error becomes smaller than the specified convergence threshold.

Illustrative Example

For the matrix

$$X = \begin{pmatrix} \text{NA} & 0.3 & 2.0 & \text{NA} \\ 8.0 & 1.3 & \text{NA} & \text{NA} \\ 11.4 & 2.3 & 2.8 & 0.5 \\ \text{NA} & \text{NA} & 2.8 & 0.7 \end{pmatrix},$$

the PCA imputation algorithm was applied using different numbers of principal components.

With one principal component, the algorithm converged after 59 iterations and produced approximately

$$\hat{X}_1 = \begin{pmatrix} 5.971 & 0.300 & 2.000 & 0.322 \\ 8.000 & 1.300 & 2.102 & 0.430 \\ 11.400 & 2.300 & 2.800 & 0.500 \\ 10.748 & 1.798 & 2.800 & 0.700 \end{pmatrix}.$$

With two principal components, the algorithm converged after 78 iterations and produced approximately

$$\hat{X}_2 = \begin{pmatrix} 9.333 & 0.300 & 2.000 & 1.271 \\ 8.000 & 1.300 & 1.915 & 0.523 \\ 11.400 & 2.300 & 2.800 & 0.500 \\ 11.546 & 1.985 & 2.800 & 0.700 \end{pmatrix}.$$

Application to the Boston Dataset

The Boston dataset was standardized before introducing missing values. Missing values were then generated randomly at three different levels:

$$5\%, \quad 10\%, \quad 20\%.$$

For each missing-value percentage, PCA imputation was performed using between 1 and 8 principal components.

The quality of the imputed values was evaluated by calculating the Pearson correlation coefficient between the original values and the values estimated by the PCA imputation procedure:

$$r = \text{corr}(X_{\text{original}}, X_{\text{imputed}}).$$

The reconstruction error was also monitored over the iterations to verify convergence of the algorithm.

Conclusion

The results show that the choice of the number of principal components affects the quality of the imputed values. In the experiments, using

$$\boxed{M = 3}$$

principal components gave the highest correlation between the real values and the imputed values.

Therefore, three principal components provided the closest reconstruction of the missing observations among the numbers of components considered. This indicates that, for this dataset and these missing-value experiments, PCA with three components can be used to obtain imputed values that are relatively close to the original values.